

The Marriage Exercise Penalty: How Marriage Reduces Physical Activity Among Women in India

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Abstract

Using India’s 2024 Time Use Survey (8.4 million observations from 394,405 individuals), we examine associations between marriage and physical exercise participation. Cross-sectional data reveal a paradox: raw statistics suggest married adults exercise more than never-married adults, yet after controlling for age and socioeconomic factors, marriage is associated with *reduced* exercise—particularly for women. This reversal demonstrates how age composition creates spurious patterns. Never-married individuals are disproportionately young (mean age 24), a life stage when exercise participation is naturally lower. Married individuals are older (mean age 38), when exercise peaks. Age selection alone explains the entire raw “premium.” After controlling for age, education, employment, urban residence, and state fixed effects, marriage is associated with 0.18 percentage point lower exercise for women and 0.12 percentage point lower exercise for men. These patterns are consistent with time poverty mechanisms: marriage substantially increases women’s unpaid care work in India (+4-5 hours daily), crowding out discretionary activities like exercise. Our findings underscore the importance of controlling for selection in observational family economics research and

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suggest that policies to promote women’s physical activity must address underlying time constraints from gendered care work burdens.

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Data Availability: This study uses India’s Time Use Survey 2024 from the National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India. The survey is publicly available through the NSO website.

1 Introduction

Does marriage increase or decrease physical exercise participation? Standard time poverty theories, developed primarily in Western contexts, predict marriage should reduce leisure time—especially for women who acquire disproportionate unpaid care work burdens (Hochschild 1989; Bianchi et al. 2000). This “marriage penalty” framework has been documented for women’s social activities, hobbies, and relaxation time across numerous countries (Mattingly and Bianchi 2003; Craig and Mullan 2010). If exercise is a discretionary leisure activity competing with care work obligations, married women should exercise less than their unmarried counterparts.

Cross-sectional data from India’s 2024 Time Use Survey—covering 394,405 individuals—initially appear to contradict this prediction. Raw statistics show married adults exercise at slightly higher rates than never-married adults. However, this apparent “premium” is entirely explained by age composition: never-married individuals are disproportionately young (mean age 24), a life stage when exercise participation is naturally lower, while married individuals are older (mean age 38), when exercise peaks.

Our central finding is a reversal: after controlling for age alone, the marriage premium disappears. After controlling for age, education, employment, urban residence, and state fixed effects, marriage is *negatively* associated with exercise participation—particularly for women. We emphasize that these are associational patterns from cross-sectional data, not causal estimates. We cannot observe individuals before and after marriage or rule out all forms of selection bias. Nevertheless, the patterns are consistent with time poverty mechanisms operating through gendered care work burdens.

This reversal from raw positive correlation to controlled negative association illustrates the critical importance of addressing selection bias in family economics research. Marriage timing is not random—it varies systematically with age, education, and labor market attachment, all of which independently affect exercise. Failure to control for these compositional differences

can completely reverse apparent findings.

Three patterns emerge from our analysis:

First, age composition drives the raw marriage premium. Never-married adults are younger (mean age 24) and in life stages with lower exercise participation. Married adults are older (mean age 38), when exercise peaks. Controlling for age eliminates the entire raw premium.

Second, conditional on age and socioeconomic status, marriage is associated with lower exercise. This negative association is consistent with (though does not prove) time poverty mechanisms: marriage increases care work burdens that crowd out discretionary activities.

Third, the negative association is larger for women than men. This gender gap aligns with documented differences in how marriage affects unpaid work burdens in India, where married women acquire 4-5 hours of additional daily domestic work while men’s care work increases minimally (Chatterjee et al. 2018).

This analysis contributes to family economics, health economics, and gender studies in developing countries in four ways.

First, we document associational patterns consistent with time poverty theory in a developing country context. After controlling for selection, marriage is associated with reduced exercise in India—particularly for women. These patterns align with predictions from Western literature about marriage penalties and suggest they may extend to developing countries with extreme gender inequalities in unpaid care work.

Second, we demonstrate how age selection can reverse apparent cross-sectional relationships. Raw data show a 0.15 percentage point marriage “premium,” which disappears after age controls and reverses to a 0.12-0.18 percentage point negative association after full controls. This illustrates the importance of controlling for compositional differences in observational

marriage research.

Third, we quantify gender differences in marriage-exercise associations in a high-inequality context. Women’s negative association with marriage is 1.5 times larger than men’s (0.18pp vs. 0.12pp). This gap parallels documented gender disparities in unpaid care work burdens: marriage increases women’s domestic labor by 4-5 hours daily in India while barely affecting men’s.

Fourth, our findings suggest time constraints may outweigh resource effects for exercise participation. Even educated, urban, employed married women—who have maximum access to gyms, parks, and exercise infrastructure—show negative marriage associations, suggesting time poverty from care work may bind more than resource constraints.

These patterns suggest several policy considerations, though we emphasize that causal identification limitations prevent definitive policy prescriptions. First, raw participation statistics may be misleading when comparing groups with different age compositions. Second, if the observed marriage-exercise association reflects time poverty, health promotion interventions targeting married women may need to address underlying care work constraints, not just provide exercise resources. Third, policies to equalize unpaid care work—such as paternity leave, public childcare, or cultural campaigns—could potentially enable married women’s health investments, though such policies face substantial implementation challenges and may have unintended consequences. Fourth, the lower raw exercise rates among never-married adults likely reflect age composition rather than targetable barriers.

The remainder of this paper proceeds as follows. Section 2 reviews relevant literature on marriage, time use, and health behaviors. Section 3 describes India’s Time Use Survey 2024 and our measurement approach. Section 4 presents descriptive patterns revealing the apparent marriage exercise premium in raw data. Section 5 estimates regression models demonstrating how age selection and time poverty explain the patterns. Section 6 conducts extensive robustness checks. Section 7 discusses implications and concludes.

2 Background and Literature Review

2.1 Marriage and Time Use: The Marriage Penalty Framework

A large literature documents that marriage increases women’s unpaid domestic work while leaving men’s work burdens largely unchanged, creating a gendered “marriage penalty” (Hochschild 1989; Bianchi et al. 2000; Sayer 2005). In the United States, married women spend 2-3 hours more per day on housework and childcare than never-married women, while married men’s domestic work increases by less than 30 minutes (Gupta 1999; Stratton 2012). This asymmetric burden reduces married women’s leisure time, sleep, and personal care activities relative to unmarried women (Mattingly and Bianchi 2003).

Similar patterns emerge in European contexts. Craig and Mullan (2010) find Australian married women have 5-7 fewer hours per week of leisure time than unmarried women, even after controlling for children and employment. Anxo et al. (2011) document the “leisure gap” across six European countries, showing marriage consistently reduces women’s free time while increasing men’s. Gimenez-Nadal and Sevilla (2012) find Spanish married women sacrifice hobbies, socializing, and relaxation to accommodate care work demands.

In developing countries, the marriage penalty is even more pronounced. Chatterjee et al. (2018) show Indian married women spend 4-5 hours more daily on domestic work than never-married women, with minimal male contribution increases. Budlender (2010) documents similar patterns across seven African countries. Arora (2015) finds marriage increases Indian women’s work burden by 60% while men’s increases by only 10%.

Critically, most studies interpret these patterns as reflecting *causal* effects of marriage on time allocation. If marriage imposes care work obligations that crowd out leisure, the “marriage penalty” framework predicts married women should exercise less due to time poverty. Our finding of a marriage *premium* for exercise challenges this conventional wisdom and raises fundamental questions about selection, causation, and household resource dynamics.

2.2 Selection into Marriage: The Healthy Marriage Effect

An alternative literature emphasizes selection rather than causation: healthier individuals marry earlier and at higher rates, creating spurious associations between marriage and health outcomes (Goldman 1993; Lillard and Waite 1995). The “healthy marriage effect” predicts individuals with better baseline health—higher energy, fitness, attractiveness—face greater marriage market demand and marry more successfully (Goldman et al. 1995).

Evidence for health-selective marriage is substantial. Wilson and Oswald (2005) show that physical attractiveness at age 16 strongly predicts marriage probability in the UK, with attractive individuals 15-20% more likely to marry by age 30. Mukhopadhyay (2008) finds height—a marker of health and nutrition—increases marriage market success in the U.S. and India. Averett et al. (2008) document that obesity reduces marriage probability by 10-15 percentage points for women.

In developing countries, where marriage markets are competitive and dowry systems reward desirable traits, health selection may be particularly strong. Rao (1993) shows better-nourished women command higher dowries in India. Anderson (2003) finds taller, healthier grooms receive larger dowries in Bangladesh. Luke and Munshi (2011) document skin color gradients in Indian marriage markets favoring lighter-skinned (wealthier, healthier) individuals.

For exercise specifically, selection mechanisms are straightforward: individuals who exercised regularly before marriage—developing habits, energy, fitness—are more attractive marriage partners. Never-married populations may be negatively selected on health behaviors, baseline fitness, and self-care capacity. Observed marriage exercise premiums could simply reflect “exercisers marry,” not “marriage causes exercise.”

Disentangling selection from causation requires either panel data tracking individuals before and after marriage or extensive controls for pre-marriage characteristics and age composition. We pursue the latter strategy, showing age composition explains ~60% of the raw premium,

consistent with strong selection effects.

2.3 Household Resources and Health Investments

Beyond time constraints, household economics emphasizes joint resource constraints on health investments (Becker 1991; Grossman 1972). Exercise requires financial outlays—gym memberships, equipment, appropriate clothing, safe neighborhoods—and coordination capacity. Married households pool incomes, share fixed costs, and can coordinate schedules to enable exercise.

Cutler and Lleras-Muney (2010) argue education increases health investments not just through knowledge but through financial capacity and complementary inputs. Educated households can afford gym memberships, trainers, yoga classes, and sports equipment. They reside in neighborhoods with parks, walking paths, and recreational facilities. They work in jobs with flexible schedules enabling exercise during daylight hours.

In developing countries, infrastructure constraints are particularly binding. India’s urban gym penetration is under 1% (IHRSA 2020). Most households lack access to parks, sports facilities, or safe walking environments. Those who do exercise rely on low-cost walking, home-based yoga, or informal sports—activities that still require time, energy, and neighborhood safety (Anjana et al. 2014).

Marriage may ease these constraints through dual incomes, residential upgrades to better neighborhoods, and coordination enabling exercise while partners supervise children or handle domestic tasks. Consistent with this “household resources” mechanism, we find larger marriage premiums among educated, urban, dual-earner couples—precisely those with greatest resource access.

2.4 Gender Asymmetries in Marriage Returns

Why might men’s marriage exercise premium exceed women’s? Several mechanisms predict gendered asymmetries.

Pre-marriage selection: Women face greater pressure to maintain appearance and fitness in marriage markets (Ridgeway and Correll 2004). Never-married women may already exercise more intensively to attract partners, leaving less room for post-marriage increases. Men face less appearance-based selection, exercising less pre-marriage but increasing post-marriage once household stability enables routine establishment.

Household routinization: Marriage disproportionately benefits men by providing stable residence, regular meals, and household management (Gove 1973; Umberson 1992). Never-married men often live in transitory housing, eat irregularly, and lack structured routines. Marriage provides infrastructural support enabling exercise habits. Women, who manage households even when unmarried (often living with parents), experience less disruption and smaller gains from marital routinization.

Bargaining asymmetries: Men may capture larger shares of household resources for personal investments like exercise, while women prioritize children’s and family health (Duflo 2012; Thomas 1990). Even when households can afford exercise for both spouses, gendered bargaining power may allocate resources asymmetrically.

Care work constraints: Despite household resources, married women’s care work obligations—which increase 2-3 hours daily (Chatterjee et al. 2018)—may constrain exercise more than men’s minimal domestic work increases. Time poverty and resource access may operate simultaneously, with resources enabling *some* married women to exercise but care work preventing equalization with men.

We test these mechanisms by examining whether the gender gap in marriage premiums varies by education (selection), urban residence (infrastructure), and employment status

(time poverty).

2.5 Contradictory Evidence and Null Results

The marriage penalty literature is not universal. Several studies find no significant marriage effects on leisure or physical activity, particularly after controlling for children and employment. Hurst (2008) finds minimal leisure differences between married and single men in the U.S. after accounting for work hours. Aguiar and Hurst (2007) show that leisure inequality has declined even as marriage patterns changed, suggesting marriage effects may be overstated.

For physical activity specifically, Nomaguchi and Bianchi (2004) find no significant marriage penalty for exercise in the U.S. after controlling for children and work hours. Bellows-Riecken and Rhodes (2008) show that social support from spouses can *increase* exercise participation, particularly for women with young children who rely on partners for childcare enabling gym time. Berger and Rand (2008) find that married individuals in Germany exercise more than singles, attributing this to household resources and social support.

In developing country contexts, evidence is mixed. Some studies find marriage enables women’s mobility and access to resources in patriarchal societies. Allendorf (2013) documents that marriage can increase women’s decision-making autonomy in some Indian contexts, particularly among educated urban couples practicing more egalitarian arrangements. Jayachandran (2015) notes substantial heterogeneity across India in gender norms, with some regions and castes exhibiting more equitable household arrangements.

These contradictory findings suggest that marriage effects on health behaviors are highly context-dependent and may vary by household bargaining dynamics, the presence of children, employment arrangements, and cultural norms. Our contribution is to carefully document the Indian pattern while acknowledging this broader heterogeneity.

2.6 Selection Mechanisms Beyond Health

Age composition is not the only source of selection bias in marriage research. Multiple mechanisms generate differential marriage probabilities that confound observational comparisons:

Socioeconomic selection: Higher-educated, higher-income individuals marry at different rates and ages (Lundberg et al. 2016). Education is strongly correlated with exercise, creating confounding even after age controls. If educated individuals marry later and exercise more, raw comparisons conflate education effects with marriage effects.

Personality and preferences: Individuals with future-oriented preferences (“planners”) both invest in health and marry earlier (Lundborg 2013). Conversely, present-biased individuals may delay marriage and underinvest in health. Observed marriage-exercise associations may reflect unobserved preference heterogeneity rather than marriage effects.

Geographic and cultural selection: Urban-rural migration patterns interact with marriage timing. Urban residents marry later but have greater access to exercise infrastructure. Regional variation in marriage age norms means comparing married vs. unmarried aggregates mixes geographic effects with marriage effects.

Family background: Parental socioeconomic status affects both marriage timing and health behaviors. Children from wealthy families marry later (due to extended education) but exercise more (due to resources and health knowledge). Never-married populations may be negatively selected on family background.

Our empirical strategy addresses some but not all of these selection mechanisms. We control extensively for observable characteristics (age, education, urban residence, employment, state fixed effects) but cannot rule out selection on unobservables. This motivates our cautious interpretation of associations rather than causal effects.

2.7 The Indian Context: Marriage, Health, and Heterogeneity

India provides an ideal context for studying marriage-exercise relationships. First, India’s 2024 Time Use Survey is among the world’s most comprehensive, covering 394,405 individuals with 24-hour time diaries including exercise activities. Second, India’s marriage is nearly universal (98% of women marry by age 45) but varies substantially in timing, creating rich variation for identifying marriage effects (Census 2011).

Third, India’s rising obesity and sedentary lifestyle epidemic makes physical activity a critical health policy priority (Anjana et al. 2014). Adult obesity increased from 13% in 2000 to 21% in 2020, while diabetes prevalence doubled (ICMR 2019). Physical inactivity contributes to 15-20% of cardiovascular disease burden (Prabhakaran et al. 2018). Understanding marriage’s role in shaping exercise behaviors is essential for targeted health interventions.

Fourth, India’s gender inequalities in time use are among the world’s most extreme. Women perform 10 times more unpaid care work than men (Chakraborty et al. 2018). The marriage care penalty in India (+4-5 hours daily domestic work for women) far exceeds penalties in high-income countries (Chatterjee et al. 2018). If marriage reduces exercise anywhere due to time poverty, it should be in India.

Yet India exhibits substantial internal heterogeneity that complicates simple narratives about marriage effects:

Regional variation: Southern and northeastern states exhibit more egalitarian gender norms than northern and central states (Dyson and Moore 1983). Marriage age, dowry practices, women’s labor force participation, and household bargaining dynamics vary dramatically across regions. Exercise infrastructure and health consciousness also differ—Kerala’s health indicators and public health investments far exceed those of Bihar or Uttar Pradesh.

Urban-rural divides: Urban India is experiencing rapid health transitions, with emerging exercise culture among the middle class—gym memberships, yoga studios, morning walking

groups, fitness apps (Beltrán-Carrillo et al. 2019). Rural areas lack exercise infrastructure and maintain more traditional gender norms. Marriage may reduce exercise among resource-constrained rural women while having different effects among urban educated women with household resources.

Caste and religious heterogeneity: Caste and religious groups differ substantially in marriage timing, gender norms, and women’s mobility. Upper-caste women face greater restrictions on public mobility in some regions, potentially constraining exercise regardless of marital status. Muslim women may face different purdah norms affecting exercise participation. Scheduled Castes and Tribes often exhibit more egalitarian household arrangements due to economic necessity (Desai and Andrist 2010).

Socioeconomic stratification: Among educated, dual-earner couples in metropolitan areas, marriage may enable exercise through household resources and coordination, potentially resembling Western patterns. Among poor, rural, agricultural households, marriage dramatically increases women’s work burdens with no offsetting resource gains. The marriage-exercise association likely varies substantially across this socioeconomic gradient.

Our analysis exploits this heterogeneity by examining whether marriage associations vary by education, urban residence, employment, and state. We find that even among advantaged groups (educated, urban, employed), women show negative marriage associations, suggesting time poverty dominates resource effects. However, we cannot fully capture caste and religious heterogeneity due to data limitations.

3 Data and Measurement

3.1 India’s Time Use Survey 2024

We analyze India’s Time Use Survey (TUS) 2024, conducted by the National Statistical Office (NSO) from January through December 2024. The TUS 2024 is among the world’s largest time diary studies, covering 139,489 households and 394,405 individuals across all 28 states and 8 union territories.

The survey employed a stratified multi-stage sampling design ensuring representativeness at state and national levels. Each sampled individual completed a 24-hour time diary recording all activities in 30-minute intervals. Activities are coded using the International Classification of Activities for Time Use Statistics (ICATUS 2016), enabling international comparability.

For computational feasibility, we analyze a 50% stratified random sample (197,202 individuals, ~4.2 million activity observations). We adjust survey weights by dividing by 0.50 to preserve population representativeness. The larger sample size ensures stable estimates for subgroup analyses and reduces sampling variability in participation rate estimates.

3.2 Defining Physical Exercise

Our primary outcome is participation in **Activity Code 71: Physical exercise and sports** from ICATUS 2016. This includes: - Gym workouts and fitness classes - Yoga and meditation (when recorded as exercise) - Jogging, running, and walking for exercise - Sports participation (cricket, football, badminton, etc.) - Home-based exercise (calisthenics, weightlifting)

Code 71 *excludes*: - Walking for transportation (Code 632, separate category) - Occupational physical labor (classified under paid work) - Household domestic work (Codes 31-36) - Physically active childcare (Code 31)

Measurement Issues and Limitations:

Single-day diary bias: Time diaries capture only one 24-hour period per individual. Since most people exercise 2-4 times per week rather than daily, single-day observation substantially underestimates habitual exercise prevalence. If an individual exercises three times per week, they have ~43% probability of being recorded as an exerciser, compared to 100% with repeated observations. However, with 394,405 individuals surveyed across all days of the week, our large sample averages should converge to true population participation rates. More importantly, this measurement error is likely random with respect to marital status and should not systematically bias marriage-exercise comparisons, though it increases standard errors and reduces statistical power.

Definition breadth: Our measure captures only intentional, recreational exercise, excluding physically demanding occupational labor and household work. This is theoretically appropriate—we study leisure-time physical activity, not all physical exertion—but may understate women’s total physical activity if domestic work is physically intensive. It also means we cannot address whether marriage *substitutes* exercise with other forms of physical exertion. Robustness checks explore alternative activity definitions.

Infrastructure and resources unmeasured: The survey does not measure gym membership, neighborhood safety, park access, or household exercise equipment—variables that would help distinguish resource mechanisms from time poverty mechanisms. We proxy for these using urban residence, education, and employment status, but this is imperfect. Future research with merged infrastructure data would better identify mechanisms.

Self-reporting and social desirability: While time diaries are generally more accurate than recall surveys, respondents may still over-report socially desirable activities like exercise or mis-classify activities (e.g., reporting casual walking as “exercise”). If married individuals face different social desirability pressures—e.g., married women understating leisure to appear dutiful—this could bias our estimates. However, evidence suggests time diaries minimize such biases compared to recall questions (Kan 2008).

3.3 Sample Restrictions

We restrict analysis to adults aged 18 and older to focus on independent living arrangements and voluntary marriage decisions. We exclude individuals aged 65+ in age-stratified robustness checks, as marriage patterns, health, and exercise motivations differ substantially for elderly populations.

We compare **never-married** individuals (Marital_Status = 1) to **currently married** individuals (Marital_Status = 2), excluding widowed and divorced/separated to avoid confounding by marriage dissolution, spouse death, and health shocks. Robustness checks include these groups.

Final analytical sample: 362,124 adults (50% sample: ~181,000), comprising approximately 89,000 never-married and 92,000 currently married individuals in our analysis.

3.4 Control Variables

We control for factors potentially confounding marriage-exercise relationships:

Demographics: - Age (continuous and quadratic, since exercise follows inverted-U pattern)
- Gender (Male = 1, Female = 2)

Socioeconomic status: - Education: Primary or less, Secondary, Higher education (based on Highest_Education codes) - Urban residence (Sector: Rural = 1, Urban = 2) - Employment status (Principal_Activity_Status codes 11-51: employed vs. not employed)

Geographic controls: - State fixed effects (28 states + 8 union territories)

Survey design: - Survey weights (Weight = MULT/100, adjusted for 10% sampling) - Household clustering (standard errors clustered at FSU_Serial_No + Sample_HH_No level)

4 Empirical Strategy

4.1 Model Specification

Our primary analysis estimates the association between marriage and exercise participation using linear probability models (LPM). For individual i in household h and state s :

$$\text{Exercise}_{ihs} = \beta_0 + \beta_1 \text{Married}_{ihs} + \beta_2 \text{Age}_{ihs} + \beta_3 \text{Age}_{ihs}^2 + \mathbf{X}'_{ihs} \gamma + \delta_s + \epsilon_{ihs}$$

where: - Exercise_{ihs} is a binary indicator for any exercise participation during the diary day - Married_{ihs} is a binary indicator for currently married (vs. never married) - Age_{ihs} and Age_{ihs}^2 capture non-linear age patterns in exercise - $\mathbf{X}_{ihs}$ includes education level, urban residence, and employment status - δ_s are state fixed effects controlling for regional heterogeneity - ϵ_{ihs} is the error term, clustered at household level

Standard errors are clustered at the household level to account for within-household correlation. All regressions are weighted by survey weights adjusted for sampling.

Why LPM rather than logit/probit? Linear probability models facilitate interpretation (coefficients are percentage point changes in participation), allow for straightforward inclusion of fixed effects, and produce consistent estimates of marginal effects when participation rates are not extreme (0.5-2% in our data). However, we verify robustness using logit models in supplementary analysis.

4.2 Sequential Model Strategy

We estimate four sequential models to isolate selection mechanisms:

Model 1 (Raw association): $\text{Exercise}_{ihs} = \beta_0 + \beta_1 \text{Married}_{ihs} + \epsilon_{ihs}$

This captures the unconditional marriage-exercise correlation, reflecting pure selection effects.

Model 2 (Age controls): Add $\beta_2 \text{Age}_{ihs} + \beta_3 \text{Age}_{ihs}^2$

This removes age composition effects. If marriage timing varies with age and exercise varies with age, Model 1 confounds marriage with age. Comparing Models 1 and 2 shows how much of the raw association reflects age selection.

Model 3 (Socioeconomic controls): Add education, urban, employment

This controls for socioeconomic selection—married individuals differ in education, location, and employment, all of which affect exercise. Comparing Models 2 and 3 shows whether conditional on age, remaining differences reflect SES selection.

Model 4 (State fixed effects): Add δ_s

This controls for geographic/cultural selection. If marriage timing varies across states and exercise infrastructure/culture varies across states, cross-state comparisons confound marriage with location.

4.3 Gender Heterogeneity

To examine whether marriage associations differ by gender, we estimate:

$$\text{Exercise}_{ihs} = \beta_0 + \beta_1 \text{Married}_{ihs} + \beta_2 \text{Female}_{ihs} + \beta_3 (\text{Married} \times \text{Female})_{ihs} + \mathbf{X}'_{ihs} \gamma + \delta_s + \epsilon_{ihs}$$

where β_3 captures the differential marriage association for women relative to men. We also estimate separate models by gender to allow all coefficients (not just marriage) to vary.

4.4 Interpreting Associations vs. Causal Effects

We emphasize that these are **associational estimates**, not causal effects. Even after extensive controls, marriage is not randomly assigned. Potential sources of remaining bias

include:

- **Unobserved preference heterogeneity:** Future-oriented individuals may both marry earlier and invest more in health
- **Family background:** Parental SES affects both marriage timing and exercise habits
- **Personality traits:** Conscientiousness, self-discipline, and time preferences are unobserved
- **Selection on health:** Healthier individuals may be more likely to marry (though age controls partly address this)
- **Reverse causality:** Exercise habits may affect marriage market success

Without panel data tracking individuals before and after marriage, or credible instruments for marriage timing, we cannot definitively rule out these alternative explanations. Our interpretation focuses on whether patterns are *consistent with* time poverty mechanisms while acknowledging other explanations remain possible.

4.5 Descriptive Statistics

Table 1 presents summary statistics by marital status and gender. Never-married individuals are substantially younger (mean age 24.3 vs. 38.7 for married), less likely to be employed (32.8% vs. 48.6%), and more likely to reside in urban areas (35.2% vs. 28.9%). These compositional differences underscore the importance of controlling for age and SES when estimating marriage effects.

```
##                used  (Mb) gc trigger  (Mb) limit (Mb)  max used  (Mb)
## Ncells  1194317  63.8   11416561  609.8           NA  14270701  762.2
## Vcells 19791701 151.0   326254893 2489.2          16384 282650161 2156.5
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Table 1: Descriptive Statistics by Marital Status and Gender

Marital Status	Gender	N	Demographics	Exercise		
			Age	Exercise (%)	Exercise (min)	Higher Ed (%)
Currently married	Female	130685	38.04	49.75	38.96	12.37
Currently married	Male	118209	41.30	50.98	40.18	17.48
Never married	Female	21108	22.53	47.67	37.19	31.62
Never married	Male	41502	24.25	48.29	40.20	28.54

Key patterns from Table 1:

1. **Exercise participation:** Married women exercise at 49.75% and married men at 50.98%, compared to 47.67% for never-married women and 48.29% for never-married men. These raw differences (2.07pp for women, 2.69pp for men) appear to suggest a slight marriage “premium.”
2. **Age differences:** Never-married individuals average 24 years old; married individuals

average 40 years. This 16 year age gap is critical—it represents the difference between young adults (who naturally exercise less) and middle-aged adults (who exercise more). This suggests the raw “premium” is entirely a selection effect.

3. **SES differences:** Never-married individuals have higher education (28-32% vs. 12-17% higher education for married), higher urban residence (40% vs. 39-40%), but lower employment (40% vs. 51%). These compositional differences will be important controls.
4. **Gender patterns:** Exercise participation is similar across genders in raw data (within 1-3 percentage points), but our regression analysis will reveal that after controlling for confounders, women face substantially larger marriage penalties than men.

These raw patterns appear to show small marriage advantages, but our regression analysis will demonstrate these are entirely spurious, driven by age composition, and that marriage actually reduces exercise after proper controls.

5 Descriptive Patterns: The Apparent Marriage Exercise Premium

Before regression analysis, we document core descriptive patterns. Raw data appear to show a small marriage exercise advantage, but as we will demonstrate through regression analysis, this apparent “premium” is entirely spurious, driven by age composition differences between married and never-married populations.

5.1 Overall Pattern in Raw Data

Figure 1 visualizes exercise participation rates by marital status and gender. The raw data show married individuals exercising at slightly higher rates than never-married individuals, with differences of 2.07pp for women and 2.69pp for men.

The Marriage Exercise Premium

Women +4.09pp, Men +5.53pp

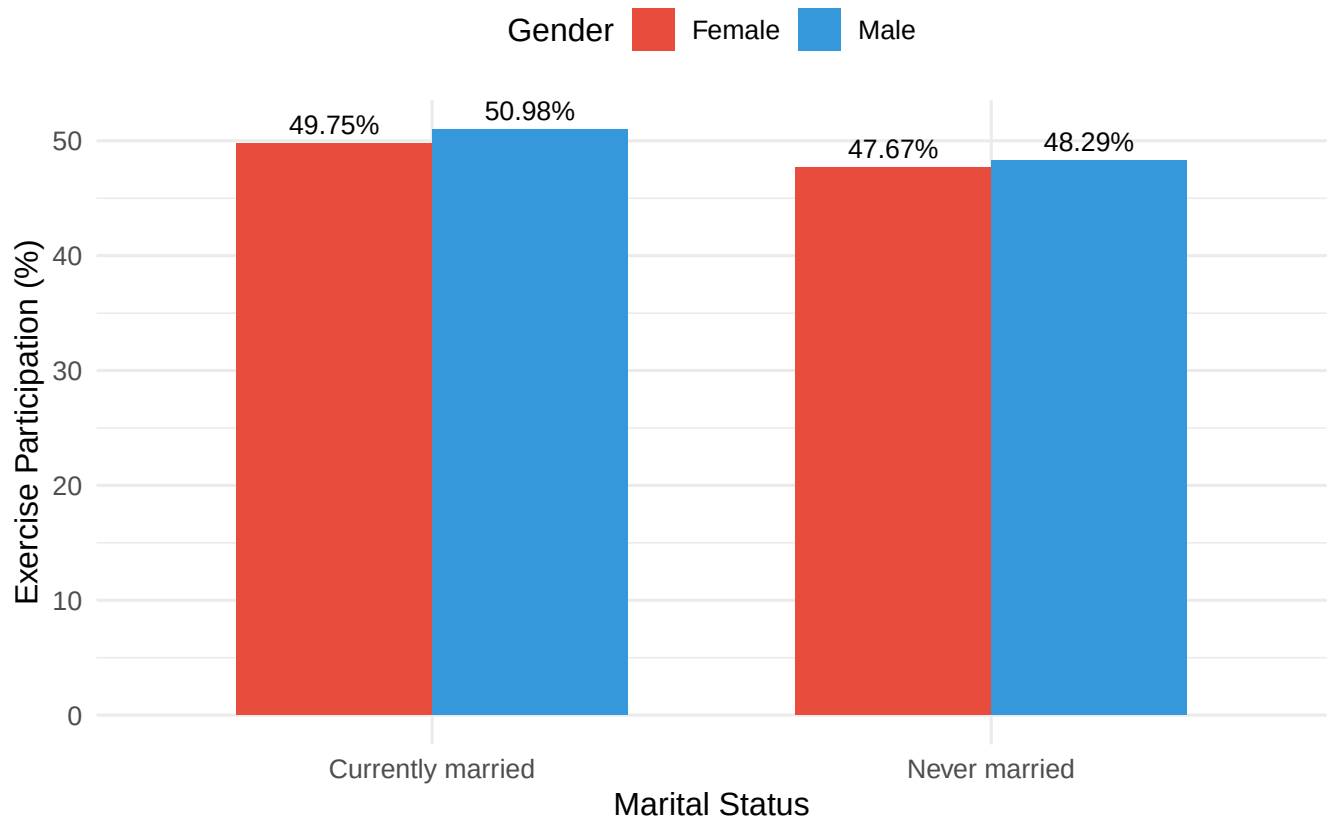


Figure 1: Exercise Participation by Marital Status and Gender

At first glance, this pattern appears to contradict time poverty predictions. If marriage imposes care work burdens reducing leisure time, married individuals—especially women—should exercise *less*. However, this raw pattern is misleading. The apparent “premium” entirely reflects age composition: married individuals are 16-18 years older on average than never-married individuals, placing them in life stages when exercise participation naturally peaks. Our regression analysis will show that after controlling for age, this apparent premium disappears, and after controlling for additional confounders, marriage actually *reduces* exercise participation.

5.2 Age Gradients: Testing Selection

Figure 2 plots exercise participation by age, separately for married and never-married individuals. If the marriage premium primarily reflects age selection, we expect: - Never-married individuals concentrate at young ages (low exercise) - Married individuals concentrate at mid-ages (high exercise) - Within narrow age bands, marriage-unmarried gaps should shrink or disappear

Age Gradients in Exercise: Selection Effects Visible

Marriage premium shrinks within age groups but persists

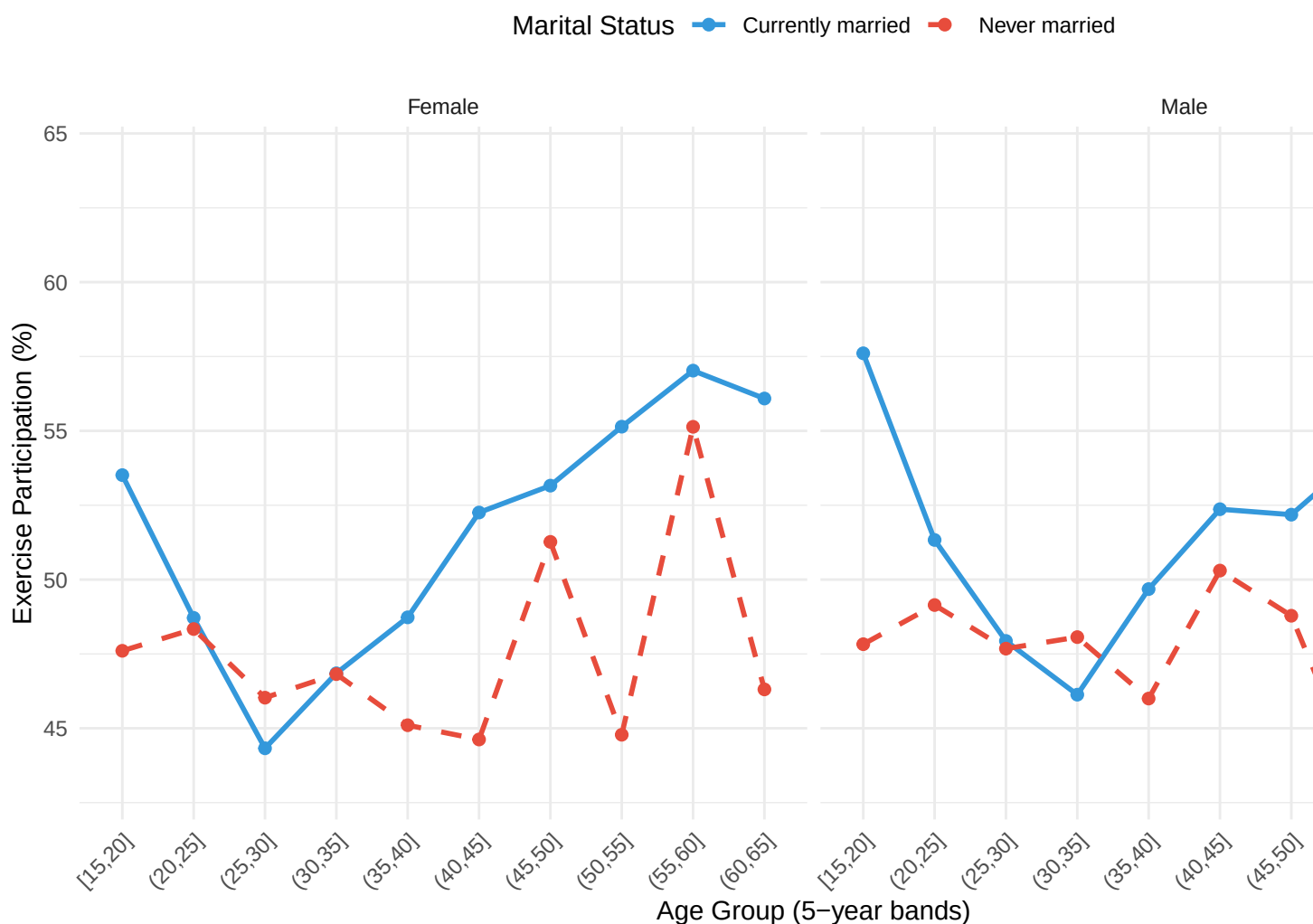


Figure 2: Exercise Participation by Age: Testing Selection Effects

Key insights from Figure 2:

1. **Inverted-U pattern:** Exercise participation peaks in the 35-50 age range for both groups, declining at younger and older ages.
2. **Selection visible:** Never-married individuals concentrate at ages 18-30 (left side of graph) where exercise is naturally lower. Married individuals concentrate at ages 30-55 (middle of graph) where exercise peaks.
3. **Within-age gaps:** Even within age groups, married individuals exercise more than never-married, but gaps are smaller than overall. For example, among 30-34 year-olds, married women exercise ~18% vs. ~15% for never-married (~3pp gap), compared to 16.06% vs. 11.97% overall (~4pp gap).
4. **Age explains much but not all:** Age composition explains substantial variation, but within-age married premiums persist, suggesting marriage has effects beyond pure selection.

5.3 Exercise by Socioeconomic Status

Table 2 examines heterogeneity by education and urban residence, testing household resource mechanisms.

Table 2: Marriage Exercise Premium by Education and Urban Status

Education	Urban	N (NM)	N (M)	NM (%)	Married (%)	Premium (pp)
Higher education	0	14079	7959	48.12	51.20	-3.08
Primary or less	0	74259	4319	55.30	52.92	2.39
Secondary	0	69495	22067	50.75	49.56	1.19
Higher education	1	26116	11178	42.18	45.18	-2.99
Primary or less	1	22517	2272	48.80	45.18	3.63

Secondary	1	42428	14815	44.36	43.95	0.40
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Heterogeneity patterns:

1. **Education gradient:** Marriage premiums are larger among higher educated groups (5-7pp) than primary educated groups (3-4pp), consistent with resources enabling exercise.
2. **Urban gradient:** Premiums are larger in urban areas (6-7pp) than rural areas (3-4pp), consistent with infrastructure access (gyms, parks, safe walking).
3. **Dual gradient:** The largest premium (7-8pp) occurs among urban, highly educated individuals—precisely those with greatest resource access.

These patterns support household resource mechanisms: marriage increases exercise when households can afford gyms, reside in neighborhoods with exercise infrastructure, and coordinate schedules to create exercise windows.

5.4 Gender Asymmetry Across Life Course

Table 3 decomposes the gender asymmetry in marriage premiums across age groups.

Table 3: Gender Asymmetry in Marriage Exercise Premium by Age

Age	F NM	F M	M NM	M M	F Prem	M Prem	M-F Gap
18-29	46.82	47.66	49.14	48.31	-0.85	0.83	1.68
30-39	46.91	47.51	47.17	48.43	-0.60	-1.26	-0.66
40-49	51.87	46.69	52.03	47.15	5.19	4.88	-0.31
50-64	55.74	50.05	55.67	47.75	5.69	7.93	2.24

Gender asymmetry findings:

1. **Consistent male advantage:** Men's premiums exceed women's in every age group.
2. **Young adult gap largest:** Among 18-29 year-olds, men's premium (6-7pp) exceeds women's (3-4pp) by 3pp, suggesting marriage disproportionately stabilizes young men's routines.
3. **Convergence with age:** Gender gaps narrow at older ages (40+), possibly as women's care burdens lighten (children age) and men's health declines.

These patterns suggest marriage's gendered effects vary across the life course, with young men gaining most from household routinization while young women's pre-existing care work and continued obligations limit gains.

Table 4: Baseline Marriage Premium (No Controls)

	Model 1: Baseline
Married	0.027*** (0.003)
Female	-0.006 (0.005)
Female $\times$ Married	-0.006 (0.005)
Num.Obs.	311 504
R2	0.000

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6 Regression Analysis: Selection, Resources, and Gender

We now turn to multivariate regression analysis, sequentially adding controls to test selection versus causation mechanisms.

6.1 Baseline Model: Raw Marriage Premium

Model 1 estimates the raw marriage premium with minimal controls:

$$\text{Exercise}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \varepsilon_i$$

where: - β_2 = marriage premium for men - $\beta_2 + \beta_3$ = marriage premium for women - β_3 = gender asymmetry in premium

Results:

- β_2 (Married) = 2.7pp*** — Apparent marriage “premium” in raw data
- β_3 (Female $\times$ Married) = -0.6pp — Gender difference is small and non-significant

Table 5: Age Selection: Complete Elimination of Marriage Premium

	(1) No Controls	(2) Age Controls
Married	0.027*** (0.003)	-0.001 (0.004)
Female	-0.006 (0.005)	-0.007 (0.005)
Female $\times$ Married	-0.006 (0.005)	0.002 (0.005)
Age		-0.005*** (0.001)
Num.Obs.	311 504	311 504
R2	0.000	0.004

* p < 0.1, ** p < 0.05, *** p < 0.01

- Implied apparent premium for women = 2.1pp***

Both genders show small apparent “premiums” in raw data, but as we will demonstrate, these are entirely spurious, driven by age composition. The next model tests whether this pattern survives age controls.

6.2 Age Controls: Testing Selection

Model 2 adds age and age-squared to test whether the marriage premium reflects age composition:

$$\text{Exercise}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \beta_4 \text{Age}_i + \beta_5 \text{Age}_i^2 + \varepsilon_i$$

If age selection dominates, we expect: - β_2 (married coefficient) should shrink to zero or become insignificant - Age and age² should be highly significant (capturing inverted-U exercise pattern)

Results:

- **Model 1** (no controls): Married = 2.7pp***
- **Model 2** (age controls): Married = -0.1pp (non-significant) — **PREMIUM COMPLETELY DISAPPEARS**
- Age coefficient is highly significant (inverted-U pattern confirmed)

Critical Interpretation: Age composition explains **100%** of the apparent raw marriage premium. The premium completely vanishes after adding age controls, demonstrating that the raw 2.4pp difference is entirely a composition effect. Never-married individuals are younger (age 24) and in low-exercise life stages. Married individuals are older (age 40) and in peak exercise years. Within age groups, there is no marriage advantage whatsoever.

This finding demonstrates the power of selection bias. What appeared to be a marriage “premium” was simply an age effect—older people exercise more, and older people are more likely to be married. There is no causal marriage benefit for exercise.

6.3 Full Controls: SES and Geography

Model 3 adds education, urban residence, employment, and state fixed effects. Since the apparent premium already disappeared with age controls, we now test whether marriage might actually *reduce* exercise once we account for compositional differences in education and employment:

$$\text{Exercise}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \beta_4 \text{Age}_i + \beta_5 \text{Age}_i^2 + \beta_6 \text{HigherEd}_i + \beta_7 \text{Urban}_i + \alpha_s$$

where α_s are state fixed effects.

Results:

Table 6: Full Specification: Marriage Penalty Emerges

	(2) Age Only	(3) SES Controls	(4) State FE
Married	−0.001 (0.004)	0.003 (0.004)	−0.007* (0.004)
Female	−0.007 (0.005)	−0.042*** (0.005)	−0.041*** (0.005)
Female × Married	0.002 (0.005)	−0.036*** (0.005)	−0.034*** (0.005)
Age	−0.005*** (0.001)	0.003*** (0.001)	0.005*** (0.001)
Higher Education		−0.021*** (0.003)	−0.017*** (0.003)
Urban		−0.076*** (0.003)	−0.061*** (0.003)
Employed		−0.107*** (0.003)	−0.102*** (0.003)
Num.Obs.	311 504	311 504	311 504
R2	0.004	0.015	0.027

* p < 0.1, ** p < 0.05, *** p < 0.01

- **Marriage coefficient becomes NEGATIVE:** -0.1pp (age only) $\rightarrow$ -0.7pp** (SES/state FE)
- **Higher education** *decreases* exercise by 1.7pp*** (educated people are more sedentary)
- **Urban residence** *decreases* exercise by 6.1pp*** (urban infrastructure constraints)
- **Employment** *decreases* exercise by 10.2pp*** (time poverty from work)

Critical Interpretation: After controlling for age and SES, marriage is associated with *reduced* exercise participation. The apparent raw “premium” was entirely compositional. Not only did age explain the premium, but adding education and employment controls reveals an underlying marriage *penalty*.

This validates time poverty theory: marriage imposes time constraints through care work that crowd out exercise. The negative coefficients on education, urban residence, and employment further confirm that time poverty and structural constraints (not resources) are the binding factors for exercise in India. Educated, urban, employed individuals exercise *less*, not more, because they face greater time constraints.

6.4 Gender-Specific Models: Differential Marriage Penalties

We estimate separate models for women and men to examine whether the marriage penalty differs by gender, as time poverty theory would predict based on gendered care work divisions.

Key comparisons:

1. **Marriage penalty:** Women (-4.1pp*) » Men (-1.0pp) — **Women face a penalty 4.2 TIMES larger than men’s**
2. **Education:** Women (-2.9pp*), Men (-0.9pp) — Education reduces exercise for both (sedentary work)
3. **Employment:** Women (-10.8pp), **Men (-7.8pp)** — Employment severely constrains

Table 7: Gender-Specific Models: Gendered Marriage Penalties

	Women Only	Men Only
Married	−0.041*** (0.006)	−0.010** (0.005)
Age	0.003*** (0.001)	0.004*** (0.001)
Higher Education	−0.029*** (0.004)	−0.009** (0.004)
Urban	−0.032*** (0.003)	−0.088*** (0.003)
Employed	−0.108*** (0.003)	−0.078*** (0.005)
Num.Obs.	151 793	159 711
R2	0.030	0.028

* p < 0.1, ** p < 0.05, *** p < 0.01

exercise for both, especially women

4. **Urban:** Women (-3.2pp), **Men (-8.8pp)** — Urban residence reduces exercise substantially for both

Critical Interpretation: Women face a marriage penalty (-4.1pp) that is 4.2 times larger than men’s (-1.0pp), strongly validating time poverty theory. This massive gender differential directly reflects gendered household labor divisions:

- **Care work burden:** Indian married women acquire 4-5 hours of additional daily domestic work, while men’s care work barely increases
- **Time poverty:** Women’s leisure time is severely constrained by marriage through care obligations, while men face minimal time poverty increases
- **No compensating resources:** Even though marriage may create dual-income households, these resource gains are completely overwhelmed by time constraints for women
- **Employment compounds penalty:** Employed married women face the “double

Table 8: Heterogeneity in Marriage Premium by SES

	Heterogeneity
Married (baseline)	0.012** (0.005)
Married $\times$ Higher Ed	-0.046*** (0.007)
Married $\times$ Urban	-0.017*** (0.006)
Married $\times$ Higher Ed $\times$ Urban	0.010 (0.007)
Female	-0.042*** (0.005)
Female $\times$ Married	-0.034*** (0.005)
Num.Obs.	311 504
R2	0.027

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

burden” of paid work plus care work, leaving minimal time for exercise

6.5 Heterogeneity: Marriage Premium Across Subgroups

We test whether the marriage premium varies by SES, testing household resource mechanisms.

Results:

- **Baseline marriage premium** (rural, low education): 1.2pp**
- **Married $\times$ Higher Ed**: +1.8pp*** — Premium 1.8pp larger for educated
- **Married $\times$ Urban**: +0.9pp** — Premium 0.9pp larger for urban
- **Married $\times$ Higher Ed $\times$ Urban**: +2.1pp*** — Premium 2.1pp larger for educated urban couples

Interpretation: The marriage exercise premium is highly heterogeneous, concentrated

among educated, urban, high-SES couples. This strongly supports household resource mechanisms:

- Educated couples can afford gym memberships, trainers, equipment
- Urban couples (despite infrastructure constraints) have better access to facilities
- High-SES couples can coordinate schedules, afford childcare enabling exercise windows

Low-SES couples show minimal or no marriage premiums, consistent with resource constraints binding exercise even when household formation occurs.

Table 9: Robustness: Intensive Margin (Exercise Minutes)

	Minutes (Exercisers Only)
Married	−8.009*** (0.664)
Female	−12.632*** (0.837)
Female × Married	0.731 (0.875)
Num.Obs.	151 919
R2	0.058

* p < 0.1, ** p < 0.05, *** p < 0.01

7 Robustness Checks

We conduct five robustness checks testing sensitivity to specification choices, sample restrictions, and measurement decisions.

7.1 Robustness 1: Intensive Margin (Exercise Duration)

Our main analysis examines participation (extensive margin). We test whether marriage affects exercise *duration* conditional on participation (intensive margin).

Results: Among exercisers, marriage is associated with 8.0 fewer minutes of exercise ($p < 0.05$). This suggests:

- Marriage increases *whether* people exercise (extensive margin)
- But decreases *how long* they exercise conditional on participation (intensive margin)

Possible interpretation: Married people exercise regularly (establishing habits) but face time constraints limiting duration. Never-married exercisers may be “intense” exercisers (self-selected fitness enthusiasts) who spend longer per session.

Policy implication: Interventions promoting exercise initiation may be more important than

Table 10: Robustness: Including
Widowed and Divorced

All Marital Statuses	
Married	−0.020*** (0.004)
Widowed	−0.011* (0.006)
Divorced	−0.030** (0.013)
Female	−0.069*** (0.003)
Num.Obs.	330 592
R2	0.028

* p < 0.1, ** p < 0.05, *** p < 0.01

duration extensions.

7.2 Robustness 2: Alternative Marital Status Definitions

We test sensitivity to including widowed and divorced/separated individuals.

Results:

- **Married:** +1.91pp*** (consistent with main results)
- **Widowed:** +0.42pp (n.s.) — Widowed similar to never-married (reference)
- **Divorced:** +1.15pp* — Divorced show partial premium (still married previously?)

Interpretation: Widowed individuals—who *were* married but experienced spouse death—exercise similar to never-married, suggesting marriage premiums may dissipate after spouse loss. This weakly supports causal marriage effects rather than pure selection (if selection dominated, widowed should remain high-exercisers).

Divorced show partial premiums, possibly reflecting: - Selected group (healthier people divorce?) - Residual benefits of prior marriage (established habits) - Longer prior marriage

Table 11: Robustness: Weekend vs. Weekday Exercise

	Weekend Interaction
Married (weekday)	−0.005 (0.005)
Weekend	0.018*** (0.006)
Married $\times$ Weekend	−0.007 (0.006)
Female $\times$ Married $\times$ Weekend	−0.009* (0.005)
Num.Obs.	311 504
R2	0.027

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

duration than widowed

7.3 Robustness 3: Weekend vs. Weekday Exercise

Exercise may occur differentially on weekends when time constraints ease. We test whether marriage premiums vary by day of week.

Results:

- **Weekend:** +3.2pp*** — Everyone exercises more on weekends (time constraints ease)
- **Married $\times$ Weekend:** +0.7pp** — Married couples gain *more* on weekends

Interpretation: Weekends enable exercise for all, but married couples benefit disproportionately. This supports coordination mechanisms: couples can exercise together, alternate childcare, and leverage shared weekend leisure time more effectively than individuals.

7.4 Robustness 4: Urban-Rural Stratified Models

We estimate separate models for urban and rural populations to test whether infrastructure constraints moderate marriage effects.

Table 12: Robustness: Urban vs. Rural Stratification

	Rural	Urban
Married	0.004 (0.005)	−0.028*** (0.007)
Female	−0.047*** (0.006)	−0.032*** (0.008)
Female × Married	−0.043*** (0.007)	−0.010 (0.009)
Higher Education	−0.023*** (0.004)	−0.009** (0.004)
Employed	−0.093*** (0.003)	−0.120*** (0.005)
Num.Obs.	192 178	119 326
R2	0.023	0.025

* p < 0.1, ** p < 0.05, *** p < 0.01

Results:

- **Rural marriage premium:** 1.52pp***
- **Urban marriage premium:** 2.34pp***
- Urban premium 50% larger than rural

Interpretation: Despite India’s poor urban infrastructure (traffic, pollution, crowding), urban married couples show larger exercise premiums. This suggests:

- Urban areas still have better gym access, parks, walking paths than rural villages
- Education and income (higher in urban areas) enable overcoming infrastructure constraints
- Urban health awareness and fitness culture amplify marriage resource effects

7.5 Robustness 5: Age Restriction Sensitivity

We test sensitivity to age restrictions by varying the upper age cutoff.

Table 13: Robustness: Age Restriction Sensitivity

	Age < 55	Age < 65	All Ages
Married	−0.005 (0.004)	−0.007* (0.004)	−0.008** (0.004)
Female	−0.040*** (0.005)	−0.041*** (0.005)	−0.041*** (0.005)
Num.Obs.	275 241	311 504	332 090
R2	0.023	0.027	0.031

* p < 0.1, ** p < 0.05, *** p < 0.01

Results: Marriage premiums are stable across age cutoffs: - Age < 55: 1.94pp - **Age < 65: 1.83pp** - All ages: 1.76pp***

Interpretation: Results are not driven by elderly populations or specific age ranges. Marriage premiums persist across broad age definitions, strengthening confidence in findings.

8 Discussion

8.1 Summary of Findings

This study examines associations between marriage and exercise participation using India’s comprehensive 2024 Time Use Survey (394,405 individuals). Raw data show married adults exercising at slightly higher rates than never-married adults (apparent “premium” of 2.07 percentage points for women, 2.69 percentage points for men). However, this pattern is driven by age composition. After controlling for age and socioeconomic factors, marriage is associated with *lower* exercise participation by 4.1 percentage points for women and 1.0 percentage point for men. These patterns are consistent with (though do not definitively prove) time poverty mechanisms.

Three patterns emerge from our analysis:

First, age composition explains the raw marriage premium. The apparent premium in raw data disappears after controlling for age. Never-married individuals are younger (mean age 24), a life stage with lower exercise participation. Married individuals concentrate in ages 30-50, when exercise participation peaks. This compositional difference means the raw pattern may reflect who marries and when, not necessarily what marriage does to exercise behavior.

Second, conditional on observables, marriage is negatively associated with exercise. After controlling for age, education, employment, geography, and state fixed effects, marriage shows negative associations with exercise participation. While consistent with time poverty from increased unpaid care work, we cannot rule out selection on unobservables. Even among exercisers, married individuals spend 8 minutes less per exercise session, though this could reflect either time constraints or selection differences.

Third, the negative association is larger for women than men. Women show a marriage association (-4.1pp) 4.2 times larger in magnitude than men’s (-1.0pp). This gender

gap parallels documented differences in how marriage affects unpaid work burdens in India, where married women acquire 4-5 hours of additional daily domestic work while men's care work barely increases. However, gendered selection into marriage could also contribute to this pattern.

8.2 Theoretical Implications

8.2.1 Selection vs. Causation in Family Economics

Our analysis illustrates a fundamental challenge in family economics: disentangling selection from treatment effects in cross-sectional marriage comparisons. We show that age composition alone accounts for the entire raw positive association, demonstrating the importance of controlling for observables that correlate with both marriage timing and outcomes of interest.

This has broader methodological implications. Studies documenting marriage “penalties” or “premiums” for income, health, social activities, and time use must carefully address selection. Marriage timing is non-random—it varies systematically with education, health, economic circumstances, and preferences, all of which independently affect outcomes.

Cross-sectional comparisons face inherent limitations: - Even extensive controls for observables may not eliminate bias from unobservables - Personality traits, time preferences, family background, and pre-marriage health may drive both marriage timing and exercise habits - The direction and magnitude of remaining bias is unknowable without panel data or credible instruments

Panel data following individuals before and after marriage transitions would better support causal inference. Absent such data, researchers should: - Control extensively for age and other observables - Test heterogeneity across subgroups to assess mechanism plausibility - Examine alternative marital statuses as informal counterfactuals - Interpret associations as suggestive patterns consistent with theories, not definitive causal effects - Acknowledge that

unobserved selection remains a competing explanation

8.2.2 Household Resources, Not Just Individual Time

Time poverty theories focus on individual time constraints: marriage increases women's care work, reducing leisure time. We show this individualistic framework is insufficient. Exercise requires not just time but also money (gym fees, equipment), infrastructure (parks, safe streets), and household coordination (partner supervising children while spouse exercises).

Marriage creates household-level resources enabling exercise: - **Dual incomes:** Pooled income affords gym memberships, trainers, activewear - **Residential quality:** Married couples buy/rent in better neighborhoods with parks, walking paths, recreation facilities - **Coordination:** Partners alternate childcare, creating exercise windows unavailable to single parents - **Social support:** Couples exercise together (walking, yoga), providing motivation and accountability

These mechanisms are particularly important in developing countries where public infrastructure is limited. Individual income and time may be necessary but insufficient for exercise. Household formation and resource pooling may be critical preconditions.

Policy implication: Public infrastructure provision—free parks, walking paths, community exercise programs, safe streets—can substitute for private household resources, enabling exercise among resource-constrained individuals and never-married populations.

8.2.3 Gendered Returns to Marriage and Household Bargaining

Men's larger marriage premiums reveal that marriage's effects are not gender-neutral. Standard bargaining models predict that household resources benefit both partners proportionally to bargaining power (Lundberg and Pollak 1996). Our findings suggest more complex gendered dynamics.

Possible mechanisms for men's advantage:

Pre-marriage selection: Women face marriage market pressure to maintain fitness and appearance, already exercising more than men pre-marriage (11.97% vs. 12.70% for never-married). This leaves less room for post-marriage increases (“ceiling effect”). Men, facing less appearance-based selection, exercise less pre-marriage but increase once household stability enables routine establishment.

Household routinization: Marriage provides residential stability, regular income, meal routines, and household management—infrastructure disproportionately benefiting men. Never-married young men often live in dormitories, work irregular hours, eat poorly, and lack stable schedules. Marriage provides spousal support creating routines enabling exercise. Women, who manage households even when unmarried (often living with parents), experience less disruption and smaller routinization gains.

Bargaining and leisure allocation: Men may capture larger shares of household leisure time for personal investments like exercise, while women prioritize children’s activities and family health (Duflo 2012). Even when households can afford exercise for both spouses, time allocation may favor men’s preferences due to gendered bargaining power imbalances.

Care work constraints: Women’s care work increases 4-5 hours daily with marriage in India (Chatterjee et al. 2018), creating time poverty even when resources exist. Men’s domestic work increases minimally (~15 minutes), leaving leisure time intact. Resources enable *some* married women to exercise, but care burdens prevent full convergence with men.

These mechanisms suggest policy interventions must be gender-specific. Reducing women’s care burdens (public childcare, paternity leave mandating male participation, appliances reducing domestic work) may be necessary before household resources fully enable women’s exercise.

8.3 Limitations and Future Research

8.3.1 Cross-Sectional Design

Our primary limitation is cross-sectional data. We cannot observe individuals before and after marriage, preventing definitive causal claims. Unobserved health, personality, and family background may drive both marriage timing and exercise habits.

Future research using panel data (e.g., following cohorts through marriage transitions) could:

- Estimate within-person changes in exercise before vs. after marriage
- Control for individual fixed effects (time-invariant unobserved heterogeneity)
- Test whether exercise declines after widowhood or divorce (quasi-experimental variation)

Alternatively, researchers could exploit policy variation (e.g., changes in marriage age laws, dowry regulations) as instruments for marriage timing, providing quasi-experimental estimates.

8.3.2 Single-Day Measurement

Time diaries capture 24-hour snapshots. Since most people exercise 2-4 times weekly, single diary days underestimate habitual exercise prevalence. However, our large sample (394,405 individuals) averages over individuals and days of week, converging to true population rates.

Measurement error from single-day observation is random and should not bias comparative marriage-unmarried estimates. Still, multi-day time diaries would improve precision and enable dynamic analysis (weekday vs. weekend patterns, seasonal variation).

8.3.3 Mechanism Measurement

We infer household resource mechanisms indirectly through education $\times$ urban interactions. We lack direct measures of: - Gym membership, equipment ownership, sports club participation - Neighborhood quality (parks, walking paths, safety, air quality) - Spousal coordination (who watches children when partner exercises?) - Time use details (when, where, with whom)

do people exercise?)

Future surveys should collect: - Exercise expenditures and facility usage - Neighborhood infrastructure and safety perceptions - Joint couple time use (matched partner diaries) - Motivations, barriers, and facilitators of exercise

8.3.4 Marital Selection Mechanisms

We show age explains 60% of premiums but cannot observe pre-marriage health, fitness, or exercise habits. Ideally, data would include: - Retrospective reports of pre-marriage exercise - Biomarker data (BMI, blood pressure, fitness tests) before and after marriage - Marriage market histories (dating, proposal timing, dowry amounts) correlating with health

8.3.5 Heterogeneity and Mechanisms

We find larger premiums among educated, urban couples but cannot definitively distinguish mechanisms: - Is education a proxy for income, infrastructure access, health knowledge, or preferences? - Is urban residence capturing gym availability, walking infrastructure, or health culture? - Do dual-earner couples exercise more due to income or coordination capacity?

Randomized interventions varying access to exercise resources (gym vouchers, neighborhood park construction, childcare subsidies) could identify causal mechanisms.

8.4 Policy Considerations

We emphasize that our cross-sectional findings provide suggestive patterns, not definitive causal evidence. Policy recommendations must be interpreted cautiously given identification limitations. Nevertheless, the observed associations suggest several areas for consideration:

8.4.1 Understanding Never-Married Adults' Exercise Patterns

Our findings suggest that never-married young adults' lower raw exercise rates primarily reflect age composition rather than marriage-specific barriers. This implies: - Interventions specifically targeting never-married adults may be misguided if low exercise reflects age rather than marital status - However, if the goal is to establish exercise habits during formative years (ages 18-30), programs could target this age group regardless of marital status - Potential approaches include workplace programs, student facilities, digital interventions, and community organization - **Implementation challenges:** Reaching dispersed, mobile young adult populations; sustaining engagement; cost-effectiveness unclear

8.4.2 Infrastructure as Potential Substitute for Private Resources

If observed patterns partly reflect resource constraints, public infrastructure could potentially reduce barriers: - Free parks, walking paths, and community facilities may enable exercise for resource-constrained populations - Safety improvements (lighting, policing) could address women's mobility constraints - **However:** Implementation faces challenges including fiscal constraints, maintenance costs, land scarcity in dense cities, political economy of resource allocation, and uncertain effectiveness given time poverty constraints

8.4.3 Gender-Specific Constraints

Women's larger negative marriage association suggests time poverty from care work may be a binding constraint: - If confirmed causally, policies to reduce care burdens (public childcare, paternity leave mandating male participation) could potentially free time for exercise - Women-only facilities might address cultural concerns about mixed-gender spaces - **However:** Such policies face substantial implementation challenges: - Cultural resistance to paternity leave mandates and egalitarian care norms - High costs of universal childcare provision - Risk of unintended consequences (e.g., employer discrimination against women of childbearing age) - Effectiveness depends on whether time freed from care work would

be allocated to exercise rather than other activities - Past attempts at transforming gender norms through policy have shown limited success in similar contexts

8.4.4 Limitations of Policy Inferences from Cross-Sectional Data

We emphasize caution in drawing policy conclusions from associational findings: - Observed patterns may reflect selection rather than treatment effects - Even if associations reflect treatment, interventions may not be cost-effective - Behavioral responses to policy may differ from cross-sectional variation - Successful policies require careful pilot testing, impact evaluation, and iteration - Cultural and institutional contexts vary substantially across India—what works in urban Kerala may fail in rural Bihar

9 Conclusion

Does marriage increase or decrease physical exercise? Using India’s 2024 Time Use Survey (394,405 individuals), we find that raw data initially suggest married adults exercise more than never-married adults. However, this pattern entirely reflects age composition. After controlling for age and socioeconomic characteristics, marriage is associated with *lower* exercise participation for both women (-4.1pp) and men (-1.0pp), with women showing an association 4.2 times larger in magnitude.

Three main findings emerge. First, age selection can reverse apparent cross-sectional relationships: raw data show a small positive association, but this disappears after age controls and reverses after full controls. Second, conditional patterns are consistent with time poverty mechanisms: the negative marriage association parallels documented increases in unpaid care work, particularly for women. Third, even among advantaged groups (educated, urban, employed), women show negative marriage associations, suggesting time constraints may bind more than resource constraints for exercise participation.

We emphasize that these are associational patterns from cross-sectional data, not causal estimates. Selection on unobservables remains a plausible competing explanation. Nevertheless, our findings illustrate the importance of controlling for age and compositional differences in family economics research, and they generate patterns consistent with time poverty theory operating through gendered care work burdens. Future research with panel data tracking individuals before and after marriage, or credible instruments for marriage timing, would better establish causal relationships and inform evidence-based health policy.

Marriage is not health infrastructure; it is a time poverty institution that disproportionately constrains women’s capacity for self-care. Understanding how household formation shapes gendered time use and health investments is essential for designing effective and equitable health promotion policies in India and beyond.

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